

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of B. Tech. (CL) Examination

November-December 2012

CL 202.01 Strength of Materials

Date: 28.11.2012, Wednesday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

- Q - 1 (a)** Define pure bending. State the assumptions made in the theory of simple bending. [04]
- (b)** A shaft of 8 m length and 400 mm diameter has its one end free and other end fixed. Find the maximum torque that can be applied on the free end if the allowable rotation is limited to 4 degree. Take $C = 10^5 \text{ N/mm}^2$. [04]
- Q - 2 (a)** A solid shaft of 200 mm diameter has the same cross-sectional area as that of a hollow shaft of the same material with inside diameter of 150 mm and the maximum shear stress developed in each of them are same. Find the ratio of the power transmitted by the shafts at the same speed. [05]
- (b)** A simply supported beam 5 m long has a section symmetrical about the y-y axis as shown in **Figure 1**. The beam carries two loads of equal magnitude as shown in **Figure 1**. If the maximum compressive and tensile stresses are not to exceed 100 N/mm^2 and 120 N/mm^2 , determine the maximum value of W. [06]

OR

- Q - 2 (a)** A hollow shaft is to transmit 300 kW at 80 rpm. If the shear stress is not to exceed 60 N/mm^2 and the internal diameter is 0.6 of the external diameter, find the external and internal diameters, assuming the maximum torque is 1.4 times the average torque. [05]
- (b)** A flitched beam consists of a wooden joist 150 mm wide and 300 mm deep strengthen by a steel plate 12 mm thick and 300 mm deep on either side of the joist. If the maximum stress in the wooden joist is 7 N/mm^2 . Find the corresponding maximum stress attained in steel. Find also the moment of resistance of the section. Take $E_s = 20 E_w$. [06]
- Q - 3 Attempt any Two** [16]
- (a)** Define principal planes.
- A rectangular block of material as shown in **Figure 2** is subjected to a tensile stress of 20 N/mm^2 on one plane and a tensile stress of 10 N/mm^2 on a plane at right angle,

together with shear stresses of 10 N/mm^2 on the same planes. Find

- (i) Magnitude and direction of the principal planes
 - (ii) Magnitude and direction of maximum shear stress
- (b) What is Mohr's circle?

Draw Mohr's circle for the element subjected to the stresses as shown in **Figure 3** and find Normal stresses, Tangential stresses and Resultant stresses on a plane inclined at 55° with vertical. Also find maximum Tangential stress.

- (c) Define principal stresses.

For an element shown in **Figure 4** find

- (i) Normal stresses, Tangential stresses and Resultant stresses on a plane inclined at 30° with vertical.
- (ii) Magnitude and direction of the principal planes
- (iii) Magnitude and direction of maximum shear stress

SECTION - II

Q - 4 (a) A circular beam is of diameter 50 mm is subjected to a shearing force of 80 kN. Find maximum shear stress and draw its shear stress distribution diagram. [02]

(b) Find moment of inertia of the section as shown in **Figure 5** about its base. [05]

Q - 5 (a) Find moment of inertia of the shaded area as shown in **Figure 6** about AB axis. [07]

(b) A steel beam section as shown in **Figure 7** is subjected to a shear force of 600 kN. Determine the shear stress distribution for the beam section. [07]

OR

Q - 5 (a) Find moment of inertia of the section as shown in **Figure 8** about centroidal axis. [07]

(b) The beam section shown in **Figure 9** is subjected to a shear force of 40 kN. Sketch the shear stress distribution for the section. [07]

Q - 6 Attempt any Two [14]

(a) Draw Shear Force and Bending Moment Diagram for the beam shown in **Figure 10**.

(b) Draw Shear Force and Bending Moment Diagram for the beam shown in **Figure 11**.

(c) Draw Shear Force and Bending Moment Diagram for the beam shown in **Figure 12**.

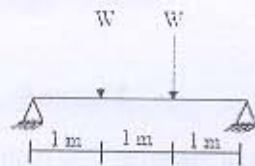
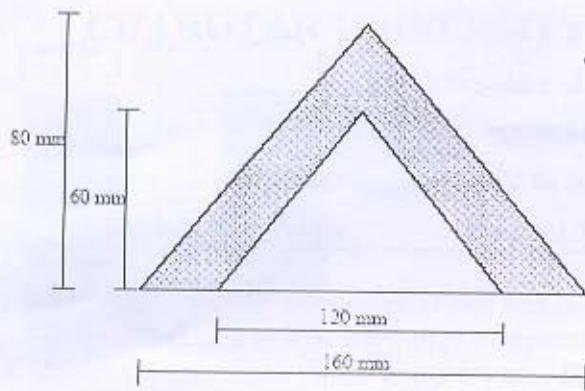


Figure 1

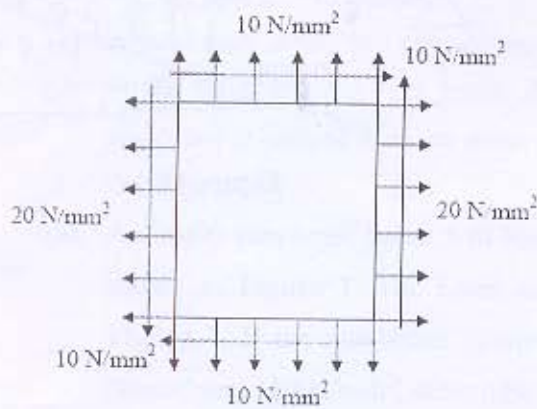


Figure 2

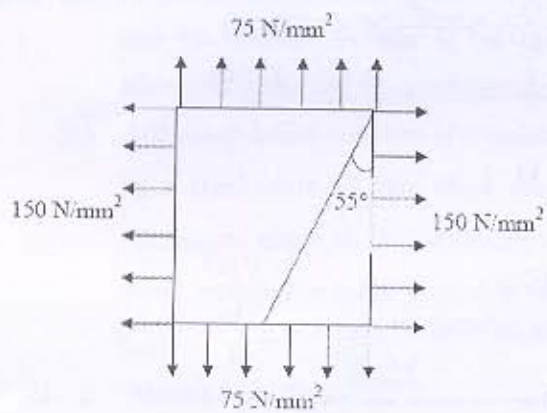


Figure 3

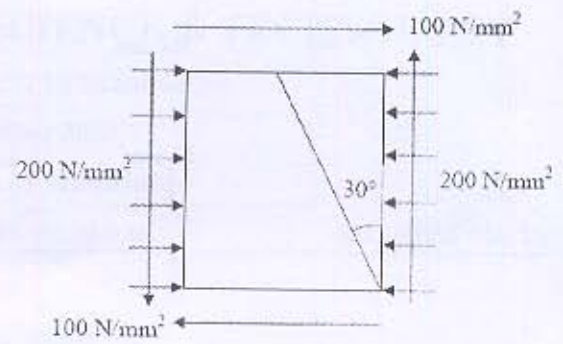


Figure 4

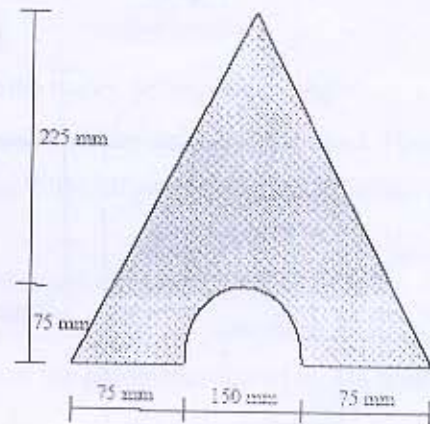


Figure 5

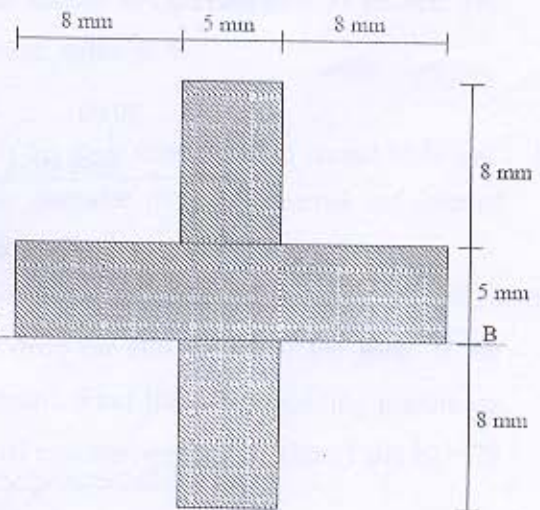


Figure 6

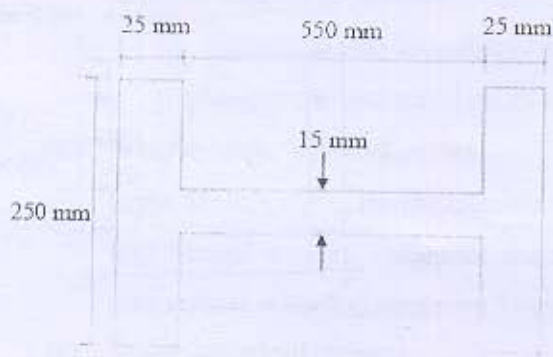


Figure 7

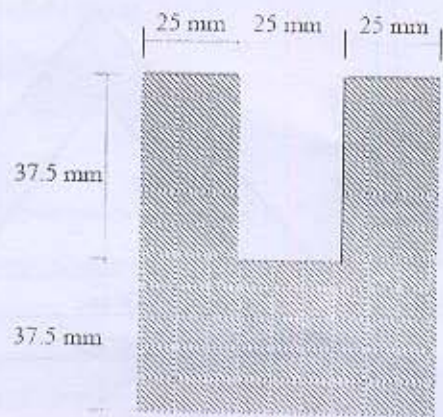


Figure 9

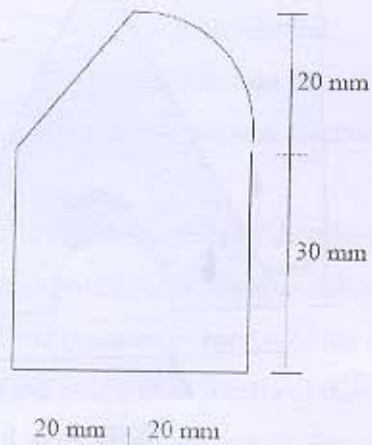


Figure 8

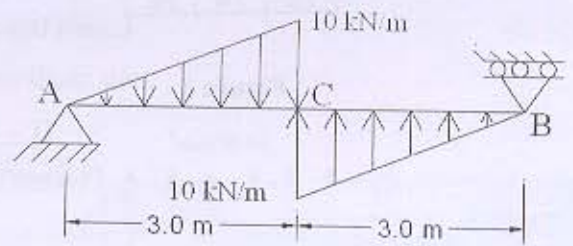


Figure 10



Figure 11

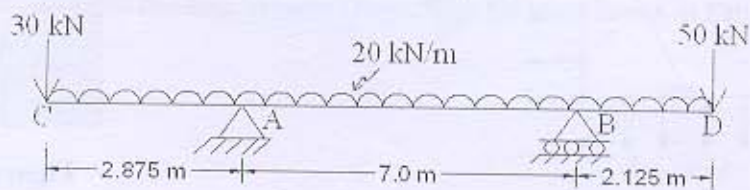


Figure 12
